Virtual memory

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Introduction

Virtual Memory **is a mechanism through which the processor can access a much larger address space than the size of the main memory actually available.**

It uses logical addresses that are then transformed into physical addresses (those that go to memory).

The mechanism aims to keep in the main memory the blocks that are expected to be used shortly by the processor, exploiting the principle of the location of the references.

The other blocks are kept in the secondary memory.

Physical and logical addresses

The processor accesses to memory generating *logical addresses*.

Physical access to physical memory is done through *physical addresses*.

The virtual memory mechanism creates a dynamic correspondence between physical and logical addresses.

The translation from logical to physical addresses, executed during the execution of the programs, is made in hardware by the *Memory Management Unit* (MMU).

Behavior

For each address generated by the processor, the MMU

- **Verifies if the requested word is physically located in the main memory**
- **If yes, the MMU produces the corresponding physical address, and the CPU can access the main memory**
- **If not (*Page Fault*) the MMU calls the Operating System, which moves the corresponding memory block into the main memory.**

Paging

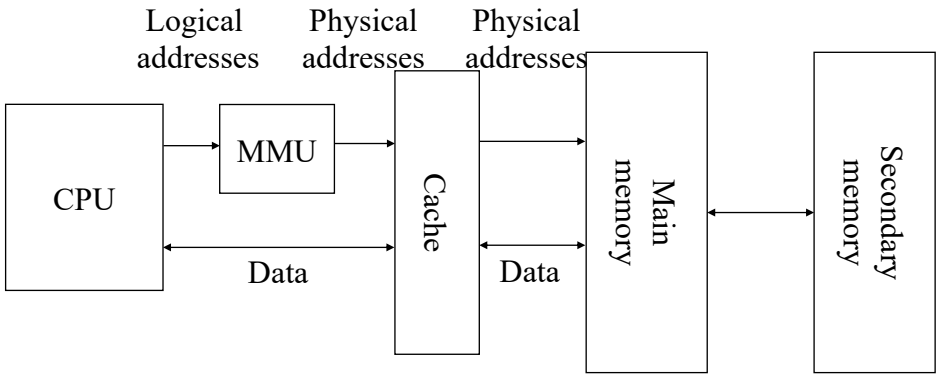
Due to the locality of references, it is convenient to adopt a mechanism such that:

- **the memory is subdivided into blocks, called *pages***
- **when the CPU accesses a data in a page that is not contained in the main memory, the entire page is transferred, usually using the DMA mechanism.**

The typical size of a page varies between 1 KB and 64 KB.

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General architecture

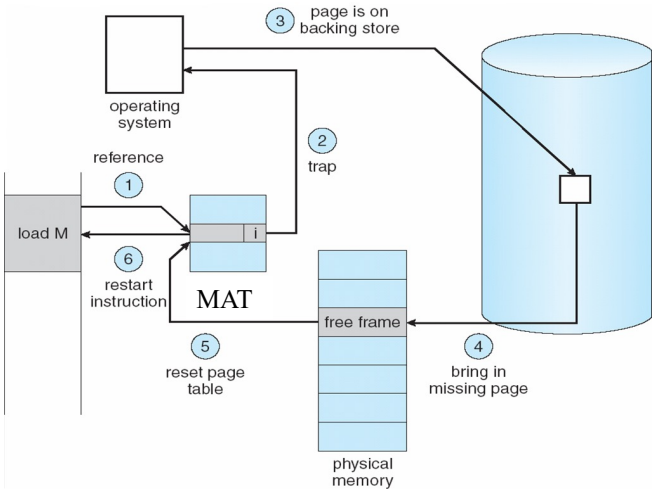


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Memory Address Table

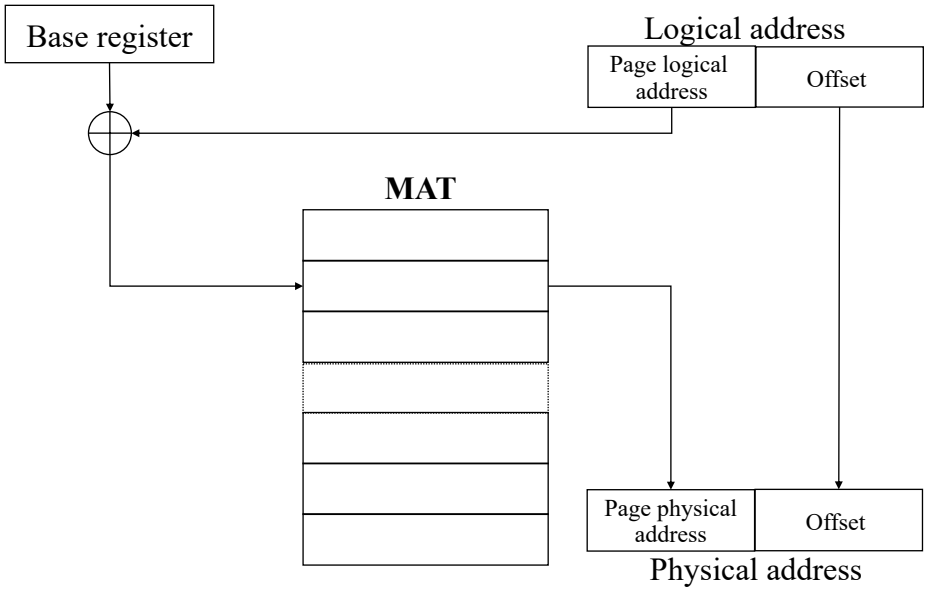
To perform transformation from physical to logic addresses, the MMU uses a table called *Memory Address Table* (MAT).
The MAT contains an entry for each virtual page.
For each entry the MAT stores the physical address where the page is in the main memory, or the page position in secondary memory.
To quickly access the correct MAT element, the MMU adds the most significant part of the address (corresponding to the virtual page number) to the contents of a register that contains the base address of the MAT.

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Address translation



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TLB

In virtual memory systems, each access to memory must transform a physical logic address, thus accessing the MAT.

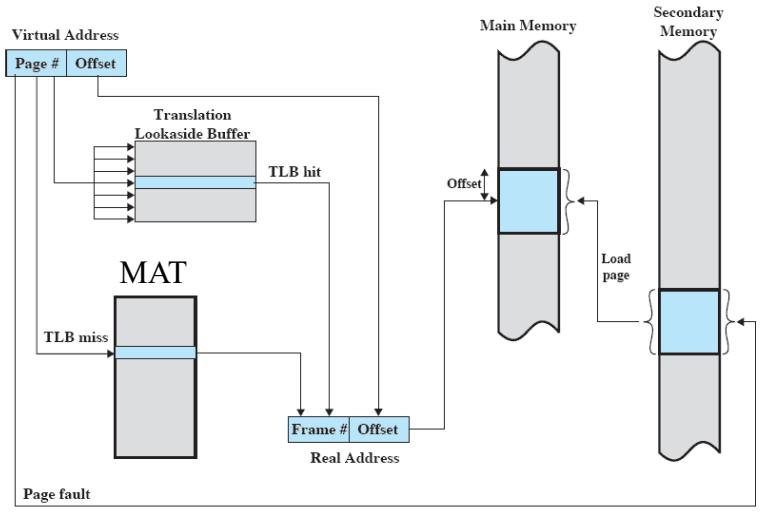
To speed up the access to the MAT, the most recently entered entries are stored in a special memory named *Translation Lookaside Buffer* (TLB).

TLB is located on board the MMU.

The TLB plays the role of a cache for the MAT.

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TLB



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i7 Cache

Characteristic	L1	L2	L3
Size	32 KB I/32 KB D	256 KB	2 MB per core
Associativity	4-way I/8-way D	8-way	16-way
Access latency	4 cycles, pipelined	10 cycles	35 cycles
Replacement scheme	Pseudo-LRU	Pseudo-LRU	Pseudo-LRU but with an ordered selection algorihtm

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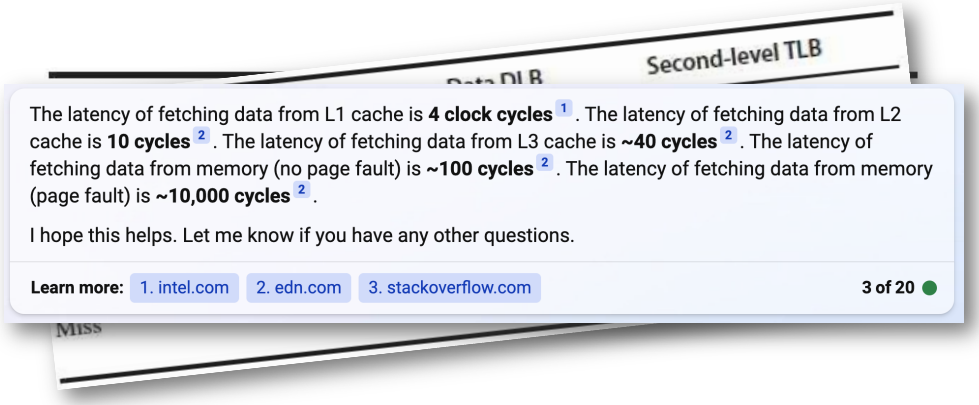
i7 TLB

Characteristic	Instruction TLB	Data DLB	Second-level TLB
Size	128	64	512
Associativity	4-way	4-way	4-way
Replacement	Pseudo-LRU	Pseudo-LRU	Pseudo-LRU
Access latency	1 cycle	1 cycle	6 cycles
Miss	7 cycles	7 cycles	Hundreds of cycles to access page table

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i7 Memory



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Advantages

The use of virtual memory offers the following advantages:

- **It makes the programs independent (from a functional point of view) from the actual system memory configuration and the real location of the data in memory**
- **it causes the programs to see a very large overall memory, with reduced access times and low cost per bit**
- **It enables efficient use of memory in multi-user environments.**

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Virtual memory vs. caches

Although there are many elements of similarity, the following features differentiate between the two mechanisms used for cache and main memory from one side, primary memory and secondary memory from the other:

- **5: 1 access time for cache, 1000: 1 for virtual memory**
- **fully HW management for caches, support of the operating system for virtual memory**
- **page sizes: 4 to 32 bytes for cache, 1 to 64KB for virtual memory.**

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Quantitative evaluation

- **The page fault management mechanisms are implemented by software from the operating system.**
- **The ratio between the access time to the secondary memory and the one in the main memory is in the order of 100 ÷ 1000.**
- **Hence, the duration of tasks resulting from miss may reach hundreds of thousands of clock periods**
- **The probability of miss must be very low (0.1% or less)**
- **The pages should be relatively large (between 1 and 64Kbyte)**
- **In the case of writing, only write-back mechanisms can be applied.**